Ashvin A. Swaminathan

Axiom Math
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Employment

2026-	Member of Technical Staff, Axiom Math
2022-26	Benjamin Peirce Fellow, Harvard University
Fall 2025	Invited Researcher, Lodha Institute of Mathematical Sciences
2022-25	NSF Postdoctoral Fellow, Harvard University Mentor: Melanie Matchett Wood

Areas of research

Number theory and algebraic geometry, with a focus on arithmetic statistics, invariant theory, Cohen–Lenstra heuristics, rational points on varieties, Diophantine equations, enumerative geometry, and AI-assisted mathematical research.

Technical skills

Experience coding in Python, Mathematica, and Matlab, as well as in subject-specific languages like sage, magma, and Macaulay2. Experience with Lean and AI-assisted research workflows.

Education

2017-22	PhD in Mathematics, Princeton University Advisor: Manjul Bhargava Thesis title: 2-Selmer groups, 2-class groups, and the arithmetic of binary forms
2016-17	AM in Physics, Harvard University
2013-17	AB in Mathematics, Harvard University *summa cum laude* & with highest departmental honors Advisors: Joe Harris & Anand Patel Senior thesis title: Inflection points in families of algebraic curves

Selected honors

Fellowships & grants

2022-25	NSF Mathematics Sciences Postdoctoral Research Fellowship
2021	US Junior Oberwolfach Fellowship
2017-22	NSF Graduate Research Fellowship
2017-21	Centennial Fellowship, Princeton University
2017-19	Paul and Daisy Soros Fellowship
2016	Barry M. Goldwater Scholarship

Awards, prizes & other recognition

2025	Hoopes Prize for Senior Thesis Advising
2021	Princeton Engineering Council Teaching Award (Math 202)
2018	AMS-MAA-SIAM Frank and Brennie Morgan Prize
2017	David B. Mumford Prize (most promising math concentrator at Harvard)
2016	Robert Fletcher Rogers Prize (best Math Table talk at Harvard)

2015-16 Harvard University Certificate of Distinction in Teaching (3 times, for Math 114, 123, 131)

Papers

Preprints and submitted articles

2026 Beyond mock modularity: elliptic corrections for higher Dyson ranks (with Claudia Alfes and Ken Ono)
Preprint. arXiv

2026 Parity of the partition function in quadratic progressions (with Ken Ono)
Preprint. arXiv

2026 On the quartic invariant of odd-degree binary forms
Preprint. arXiv github

2026 Positive proportion results for elliptic curves over $\mathbb{Q}$ with small 2-Selmer rank (with Manjul Bhargava, Wei Ho, Ari Shnidman, and Alexander Smith)
Preprint available upon request. github

2026 The average number of 3-torsion elements in class groups of orders in quadratic extensions of number fields (with Eliot Hodges)
Preprint available upon request.

2025 A parametrization of 3-class groups of quadratic rings over Dedekind domains (with Eliot Hodges)
Submitted. arXiv

2025 The second moment of the size of the 2-class group of monogenized cubic fields (with Manjul Bhargava and Arul Shankar)
Submitted. arXiv

2024 Universality theorems for zeros of random real polynomials with fixed coefficients (with Matt King)
Submitted. arXiv

2022 The second moment of the size of the 2-Selmer group of elliptic curves (with Manjul Bhargava and Arul Shankar)
Submitted. (Based on Chapter 4 of my PhD thesis) arXiv

Published articles

2024 Counting integral points on symmetric varieties with applications to arithmetic statistics (with Arul Shankar and Artane Siad)
Proceedings of the London Mathematical Society, 130(4), e70039. arXiv github

2024 Geometry-of-numbers methods in the cusp (with Arul Shankar, Artane Siad, and Ila Varma)
Algebra and Number Theory, 19(6), pp. 1099-1145. (Based on Chapter 5 of my PhD thesis) arXiv

2024 The mean number of 2-torsion elements in the class groups of cubic orders
Commentarii Mathematici Helvetici, 100(2), pp. 225-267. arXiv

2024 A positive proportion of monic odd-degree hyperelliptic curves of genus $g \geq 4$ have no unexpected quadratic points (with Jef Laga)
International Mathematics Research Notices, 2024(19), pp. 12857-12866. arXiv

2024 Most odd-degree binary forms fail to primitively represent a square
Compositio Mathematica, 160(3), pp. 481-517. (Based on Chapter 3 of my PhD thesis) arXiv

2023 A new parametrization for ideal classes in rings defined by binary forms, and applications
Journal für die reine und angewandte Mathematik, 2023(789), pp. 143-191. (Based on Chapters 1-2 of my PhD thesis) arXiv

2023 Hermite equivalence of polynomials (with Manjul Bhargava, Jan-Hendrik Evertse, Kálmán Győry, and László Remete)
Acta Arithmetica, 209, pp. 17-58, special volume in honor of Andrzej Schinzel. arXiv

2023 Inflectionary invariants for isolated complete intersection curve singularities (with Anand Patel)
Memoirs of the American Mathematical Society, 282(1397), pp. vii-99. arXiv

2021 On the EKL-degree of a Weyl cover (with Joseph Knight and Dennis Tseng)
Journal of Algebra, 565(1), pp. 64–81. arXiv

2021 Appendix to: An arithmetic count of the lines meeting four lines in $\mathbf{P}^3$ (with Borys Kadets, Padmavathi Srinivasan, Libby Taylor, and Dennis Tseng)
Transactions of the American Mathematical Society, 374(5), pp. 3447-3448. arXiv

2020 Hyperelliptic curves with maximal Galois action on the torsion points of their Jacobians (with Aaron Landesman, James Tao, and Yujie Xu)
Indiana University Mathematics Journal, 69(7), pp. 2461–2492. arXiv

2019 Surjectivity of Galois representations in rational families of abelian varieties (with Aaron Landesman, James Tao, and Yujie Xu)
Algebra and Number Theory, 13(5), pp. 995–1038. arXiv

2018 Elliptic curve variants of the least quadratic nonresidue problem and Linnik's Theorem (with Evan Chen and Peter Park)
International Journal of Number Theory, 4(1), pp. 255–288. arXiv

2017 Lifting subgroups of symplectic groups over $\mathbf{Z}/\ell\mathbf{Z}$ (with Aaron Landesman, James Tao, and Yujie Xu)
Research in Number Theory, 3(14), pp. 1–12. arXiv

2017 Permutations that destroy arithmetic progressions in elementary p-groups (with Noam D. Elkies),
Electronic Journal of Combinatorics, 24(1), pp. 1–10. arXiv

2016 On logarithmically Benford sequences (with Evan Chen and Peter Park)
Proceedings of the American Mathematical Society, 144(11), pp. 4599–4608. arXiv

2016 On arboreal Galois representations of rational functions
Journal of Algebra, 448, pp. 104–126. arXiv

2015 Linnik's Theorem for Sato-Tate laws on elliptic curves with complex multiplication (with Evan Chen and Peter Park)
Research in Number Theory, 1(1), pp. 1–11. arXiv

2014 Analysis on surreal numbers (with Simon Rubinstein-Salzedo)
Journal of Logic and Analysis, 6(5), pp. 1–39. arXiv

THESES AND EXPOSITORY ARTICLES

2022 2-Selmer groups, 2-class groups, and the arithmetic of binary forms (PhD Thesis). library record

2017 Inflection points in families of algebraic curves (Senior thesis). PDF

2016 Lie algebras and Ado's Theorem. PDF

2016 An introduction to the theory of valued fields (Junior paper). PDF

2015 On the Selberg-Erdős proof of the Prime Number Theorem. PDF

2013 Arrow's Impossibility Theorem on social choice systems. PDF

Projects

2026 Bounded gaps between primes, formalized in Lean 4 — *project lead*, Axiom Math
A complete machine-checkable Lean 4 formalization of the theorem that $p_{n+1} - p_n \leq 246$ for infinitely many n, unifying Maynard's multidimensional sieve with the enlarged-simplex refinement and numerical certificate of Polymath8b. The proofs were produced with AxiomProver and released as the open-source library PrimeGapsLib. site github
Press: *IEEE Spectrum*, "AI Used to Verify Toughest Mathematics Proof Yet" (Benjamin Skuse, 17 August 2026); *MTS* interview (19 August 2026).

Talks

2026	"Positive proportions of elliptic curves with small 2-Selmer rank" Number Theory Seminar, Boston University Workshop on Arithmetic Statistics, Centre de Recherches Mathématiques, Université de Montréal
2025-26	"Second moments for 2-Selmer structures on elliptic curves, and applications" Number Theory Seminar, Harvard University Conference on Arithmetic Statistics, Lodha Institute of Mathematical Sciences Workshop on Arithmetic Statistics of Algebraic Objects, Oberwolfach
2025	"Second moments for 2-Selmer groups and 2-class groups" Number Theory Web Seminar
2025	"Second moments for 2-class groups in families of cubic fields" Conference on Statistics of Class Groups and Number Fields, ETH Zürich
2025	"Toward secondary terms for 2-Selmer groups in special families of elliptic curves" Number Theory Seminar, Dartmouth College Number Theory Seminar, University of Basel
2025	"A positive proportion of hyperelliptic curves have no unexpected quadratic points" Algebraic Geometry Seminar, Harvard University
2024	"A positive proportion of hyperelliptic curves have no unexpected quadratic points" Algebraic Geometry Seminar, Boston College Informal Number Theory Seminar, University of Glasgow Number Theory Seminar, Tufts University Five College Number Theory Seminar, Amherst College
2024	"Finite moments for $\mathbb{Q}(i)$-ranks for elliptic curves over $\mathbb{Q}$" Lightning Talk, Workshop on Algebraic Points on Curves, ICERM
2024	"Toward secondary terms for 2-Selmer groups in families of elliptic curves" Arithmetic Statistics Seminar, Harvard University
2024	"A positive proportion of monic odd hyperelliptic curves have no unexpected quadratic points" Lightning Talk, conference on The Mordell conjecture: 100 years later
2023	"Affine symmetric spaces and class groups of cubic fields" Québec Vermont Number Theory Seminar, McGill University Conference on Algebra, Algorithms, and Arithmetic, ICMS Edinburgh Symposium on Arithmetic Geometry and its Applications, CIRM Luminy Number Theory Tea, Princeton University
2022	"On the distribution of 2-Selmer groups of hyperelliptic Jacobians" Workshop on Rational Points on Higher-Dimensional Varieties, ICMS Edinburgh
2021-23	"The second moment of the size of the 2-Selmer group of elliptic curves" Special Session on Arithmetic Statistics, JMM, Boston Number Theory Seminar, Princeton University/IAS Workshop on Explicit Methods in Number Theory, Oberwolfach
2021-22	"2-Selmer groups, 2-class groups, and the arithmetic of binary forms" Online Number Theory Seminar, University of Debrecen Number Theory Seminar, Fields Institute Palmetto Joint Arithmetic and Modularity Series (PAJAMAS) III, Invited Graduate Speaker
2020-23	"Geometry-of-numbers in the cusp, and class groups of orders in number fields" Number Theory Seminar, Duke University Conference on Arithmetic Statistics, CIRM Luminy Number Theory Seminar, MIT Montreal Online Biweekly Inter-University Seminar on Analytic Number Theory (MOBIUS ANT)
2020-22	"Most odd-degree binary forms fail to primitively represent a square" Lightning Talk, Simons Collaboration on Arithmetic Geometry, Number Theory, and Computation

Bhargava Seminar, Stanford University
Bhargava Seminar, Princeton University

2020 "Average 2-torsion in the class groups of rings associated to binary n-ic forms"
Bhargava Seminar, Princeton University
Number/Representation Theory Seminar, University of Toronto

2020 "Three lectures on the average size of 2-Selmer groups of elliptic curves"
Bhargava Seminar, Stanford University

2019 "Most hyperelliptic curves are pointless"
Bhargava Seminar, Princeton University

2017-18 "Inflectionary invariants for plane curve singularities"
Special Session on Research in Mathematics by Undergraduates, JMM, San Diego
Graduate Student Seminar, Princeton University
Algebraic Geometry Seminar, Stanford University

2017 "Surjectivity of Galois representations in rational families of abelian varieties"
Algebra, Geometry, and Number Theory Seminar, Tufts University

2017 "Surjectivity of Galois representations in rational families of abelian varieties"
Special Session on Minimal Integral Models of Algebraic Curves, JMM, Atlanta

2015-17 "Permutations that destroy arithmetic progressions in elementary p-groups"
JMM, Atlanta
Harvard Math Table

2015-16 "Elliptic curve variants of the least quadratic nonresidue problem and Linnik's Theorem"
JMM, Seattle
Harvard Math Table

2014-15 "On arboreal Galois representations of rational functions"
JMM, San Antonio
Harvard Math Table

2012-16 "Analysis on surreal numbers"
ASL Special Session, JMM, Seattle
Harvard Math Table
CMS Summer Meeting

Teaching and service

Courses taught

2024-25 Instructor, Harvard University
– Math 223br (Rational points on varieties), Q-score: 4.67/5.0.
– Math 295x (Arithmetic statistics), Q-score: 5.0/5.0.

2019 Assistant instructor, Princeton University
– Math 202 (Linear algebra with applications), Score: 4.0/5.0.

2013-17 Course assistant, Harvard University
– Math 137 (Introduction to algebraic geometry, with Peter Kronheimer), Q-score: 5.0/5.0.
– Math 131 (Topology I, with Joe Harris), Q-score: 4.5/5.0.
– Math 123 (Algebra II, with Arul Shankar), Q-score: 4.9/5.0.
– Math 114 (Analysis II, with Daniel Cristofaro-Gardiner), Q-score: 4.9/5.0.
– Math 55b (Honors real and complex analysis, with Dennis Gaitsgory), Q-score: 4.3/5.0.
– Math 55a (Honors abstract algebra, with Dennis Gaitsgory), Q-score: 4.6/5.0.

Students supervised

2024-25

Eliot Hodges — supervised senior thesis on class groups of orders in extensions of global fields
2022-24 Matt King — coauthored paper on zeros of random real polynomials

Committees and organization

2024-25 Assistant Director of Graduate Studies, Harvard Math Department
2022-25 Member, Harvard Math Department Community Committee
2022-24 Co-organizer, Harvard Number Theory Seminar

Journal refereeing

Advances in Math, *Essential Number Theory*, *Forum of Math Sigma*, *International Mathematics Research Notices*, *Involve*, *Mathematische Annalen*, *Mathematical Proceedings of the Cambridge Philosophical Society*, *Proceedings of the American Mathematical Society*, *Research in the Mathematical Sciences*, *Research in Number Theory*

Other mentoring

– Extensive experience tutoring students of all ages in math at all levels, from basic arithmetic to advanced graduate courses
– Served as volunteer tutor and class aide at Cambridge Community Learning Center to help immigrants learn math necessary to apply for jobs
– Worked with Davis International Center at Princeton to help students apply for fellowships